

EDA Summary Report

Project Title

Student Performance Analysis

Objective

The objective of this project is to analyze the Student Performance dataset using Exploratory Data Analysis (EDA). The purpose of the analysis is to understand the dataset, clean the data, calculate statistical measures, identify relationships between variables, and visualize the data using different charts.

Dataset Information

- **Dataset Name:** Student Performance Dataset
- **Source:** Kaggle
- **Number of Records:** 395
- **Number of Columns:** 33

The dataset contains information about students such as age, gender, parental education, study time, absences, and exam grades (G1, G2, and G3).

Data Cleaning

I loaded the dataset using the Pandas library. I checked the dataset for missing values and duplicate records. No major missing values were found. Duplicate records were also checked and handled before continuing the analysis. Outliers were detected using the IQR method and treated where necessary.

Statistical Analysis

I calculated the following statistical measures for the numerical columns:

- Mean
- Median
- Mode
- Variance

- Standard Deviation

These measures helped me understand the distribution and spread of the data.

Correlation Analysis

I generated a correlation matrix to study the relationship between numerical variables.

Observations

- The highest positive correlation was between **G2 and G3 (0.904868)**, which indicates that students who scored well in the second period exam also performed well in the final exam.
 - **G1 and G2 (0.852118)** and **G1 and G3 (0.801468)** also showed strong positive correlations.
 - The **failures** column had a negative correlation with the final grade (**G3 = -0.360415**), indicating that students with more previous failures generally had lower final grades.
 - Most of the remaining variables showed weak correlations with the final grade.
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Data Visualization

To better understand the dataset, I created the following visualizations:

- Bar Chart
- Line Chart
- Histogram
- Scatter Plot

These graphs helped me understand the distribution of grades and the relationship between different variables.

Key Insights

- Previous exam scores (G1 and G2) are strongly related to the final grade (G3).
- Students with more previous failures generally scored lower in the final exam.
- Most variables had only a weak relationship with the final grade.
- Data visualization made it easier to identify trends and patterns in the dataset.

Conclusion

This project helped me understand the complete process of Exploratory Data Analysis. I learned how to clean a dataset, calculate statistical measures, detect outliers, analyze correlations, and visualize data using Python libraries such as Pandas, NumPy, and Matplotlib. The project improved my understanding of how data analysis can be used to identify meaningful patterns and relationships.
